Week 2

Description and classification of sounds

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Consonants

Definition

- The sounds which are articulated with some kind of stricture, or closure, of the air stream.
- Those segments which occur at the edges of syllables, and are optional in the syllables.

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Classification

- In order to form consonants, the air-stream through the vocal organs must be obstructed in some way. Therefore, consonants can be classified according to:
  - the place where the air-stream is obstructed (***the place of articulation***), and
  - the way in which the air-stream is obstructed (***the manner of articulation***).

According to place of articulation

1. **Bilabial**:

   E.g. /p/, /b/, /m/, /w/

2. **Labio-dental**:

   E.g. /f/, /v/

3. **Inter-dental/Dental**:

   E.g. /θ/, /ð/

4. **Alveolar**:

   E.g. /t/, /d/, /s/, /z/, /l/, /n/

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According to place of articulation

5. **Alveolo-palatal/Palato-alveolar**:

   E.g. /ʃ/, /ʒ/, /tʃ/, /dʒ/, /r/

6. **Palatal**:

   E.g. /j/

7. **Velar**:

   E.g. /k/, /g/, /ŋ/

8. **Glottal**:

   E.g. /h/

According to manner of articulation

- Manner of articulation is the way in which the air-stream is obstructed or altered in the production of speech sounds.

- It describes the types of obstruction caused by the narrowing or closure of the articulators.

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According to manner of articulation

1. **Stop (Oral stop/Plosive):**
   E.g. /p/, /t/, /k/, /b/, /d/, /g/
2. **Nasal (Nasal stop):**
   E.g. /m/, /n/, /ŋ/

- **Note**: For every stop position in English, there is a nasal articulated in the same position.
  /p/ and /b/, /t/ and /d/, /k/ and /g/ vs. /m/, /n/, /ŋ/

According to manner of articulation

3. **Fricative (Spirant)**:

   E.g. /f/, /v/, /θ/, /ð/, /s/, /z/, /ʃ/, /ʒ/, /h/

- **Notes**: Fricatives are continuants consonants which means that you can continue making them as long as you have enough air in your lungs.

4. **Affricate**: A stop is immediately followed by a homorganic fricative. E.g. /tʃ/, /dʒ /

5. **Approximant (Frictionless continuant)**:
   a. **Lateral**: E.g. /l/
   b. **Retroflex**: E.g. /r/
   c. **Glide (Semi-vowel)**: E.g. /w/, /j/

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According to voicing

1. **Voiced consonants**:
   E.g. /b/, /v/, /d/, /z/

2. **Voiceless consonants**:
   e.g. /p/, /f/, /t/, /s/

English consonants' phonetic notation

- **See Table 2.1 (Brinton, 2000, p.27)**
- **The description includes the following information:**

  a. Voicing

  b. Place of articulation

  c. Manner of articulation

  E.g. /s/: voiceless alveolar fricative

  /n/: voiced alveolar nasal

/f/:	/d/:
/θ/:	/j/:
/g/:	/tʃ :
/h/:	/ŋ/:

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English consonants' Phonetic Notation

- **Answers:**

  /f/:

  /d/:

  /θ/:

  /j/:

  /g/:

  / tʃ/:

  /h/:

  /ŋ/:

Identifying English consonants

- **A description is given and you have to identify which sound is being described**

  E.g. voiced velar nasal:

  voiceless palato-alveolar fricative:

  voiced bilabial stop:

  voiced labio-dental fricative:

  voiced alveolar lateral:

  voiceless palato-alveolar affricate:

  voiced dental fricative:

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Vowels

Definition

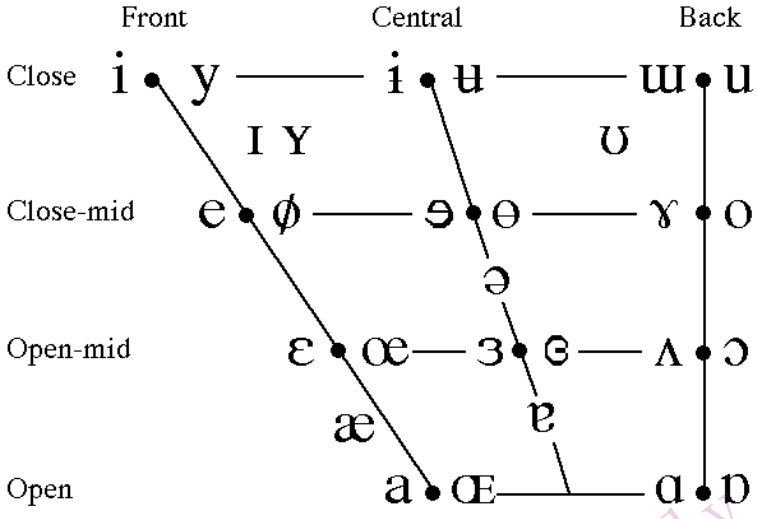
- Vowels are the sounds in the production of which none of the articulators come very close together so the passage of air-stream is relatively unobstructed and the air can get out freely.
- Vowels depend mainly on the variations in the position of the tongue. They are normally voiced.

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Cardinal vowels

- Represent the range of vowels that the human articulators can make.
- Primary cardinal vowels: most familiar to the speakers of most European languages.
- List of cardinal vowels: *http://www.phonetics.ucla.edu/course/chapter1/vowels.html*

Cardinal vowels (and more)

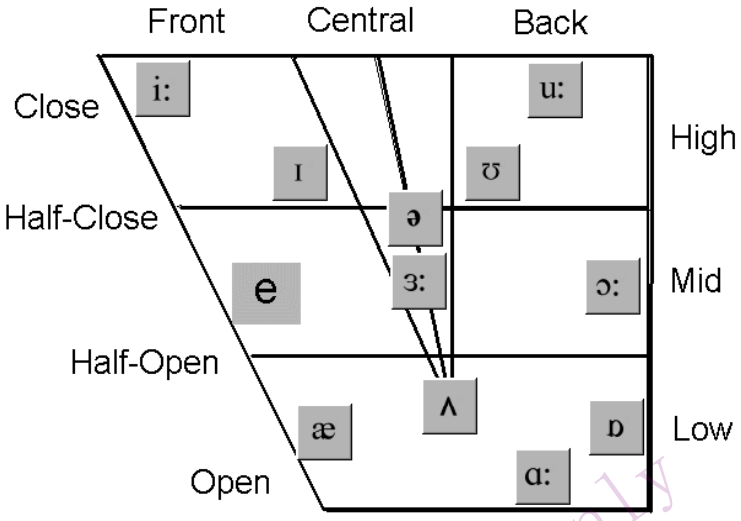


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Classification

- Vowels (monophthongs) in English can be classified according to three variables:
  a. Tongue height
  b. Part of the tongue which is raised
  c. Degree of lip rounding

Classification



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According to tongue height

1. **High vowels**:

   e.g.: /u:/, /i:/

2. **Low vowels**:

   e.g.: /ɑ:/, /æ/

3. **Mid vowels**:

   e.g.: /e/, /ɜ:/

According to the part of the tongue raised

1. **Front vowels**:

   e.g. /iː/, /æ/

2. **Back vowels**:

   e.g. /ʊ/, /ɒ/

3. **Central vowels**:

   e.g. /ə/, /ʌ/

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According to degree of lip rounding

1. **Rounded vowels**:

   E.g. /ʊ/, /ɒ/, /uː/, /ɔː/

2. **Unrounded vowels**:

   E.g. /iː/, /ɪ/, /e/

3. **Neutral vowels**:

   E.g. /ɜː/, /ə/, /ɑː/, /ʌ/, /æ/

Long and short vowels

- **Long vowels**: e.g. /i:/, /u:/
- **Short vowels**: e.g. /ɪ/, /ʊ/
- Long vowels tend to be longer than short vowels in similar contexts. The symbols consist of one single vowel plus a length mark made of two dots.

- They are different from short vowels not only in length but also in quality, resulting from differences in tongue shapes and lip positions.

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Diphthongs

- **Definition**:
  - A diphthong is a glide from one vowel to another, and the whole glide acts like one of the long simple vowels.
  - In terms of length, diphthongs are like long vowels.

    E.g. /aɪ/, /eɪ/

Diphthongs

- **Classification:**
  - Centring: ending in /ə/
  - Closing
    - Ending in /ɪ/
    - Ending in /ʊ/

- Which group do these vowels belong to?

/əʊ/	/ɔɪ/	/ʊə/	/ɪə/
/aɪ/	/eə/	/aʊ/	/eɪ/

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Describing diphthongs

- Describe the glide from the first vowel to the second.
- /aɪ/: low central to high front diphthong
- /eɪ/
- /ɔɪ/
- /eə/
- /ɪə/
- /ʊə/
- /əʊ/
- /aʊ/

English vowels phonetic notation

Describing:

- long/short
- high/low
- front/back
- rounded/unrounded.

E.g. /e/ - short mid front unrounded vowel

/u:/ - long high back rounded vowel

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Identifying vowels

- **Identification**:

E.g. long mid central unrounded vowel - …..

short low front neutral vowel - …..